

Networks — what they are

What is a network?

- A network is simply **things connected to each other** — in computing, those things are devices
- Can be as small as 2 devices (your phone and your laptop) or as big as billions of devices
- Devices include: laptops, phones, security cameras, traffic lights, farm sensors — anything technological
- Two types: **private networks** (your home, a business) and **public networks** (the Internet)

Easy way to remember it

Think of your friend group

You're all connected because of shared interests. A network is the same thing — devices connected because they need to share information. The Internet is just a massive version of that friend group, made up of millions of smaller groups all connected together.

What is the Internet?

- The Internet = one giant public network made up of millions of tiny private networks joined together
- Started as **ARPANET** in the late 1960s — funded by the US Defense Department
- The Internet we know today was created in **1989** by **Tim Berners-Lee** — he invented the **World Wide Web (WWW)**
- Before 1989 it was just a tool for sending data — after 1989 it became a place to store and share information

Memory hook: ARPANET (1960s) = the road was built. Tim Berners-Lee (1989) = the shops, houses, and signs were added. The road existed before, but nobody could really use it.

IP addresses — how devices identify themselves

What is an IP address?

- IP = **Internet Protocol**. An IP address is like a home address for a device on a network
- Made up of **4 numbers** separated by dots — called **octets**. Each number is between 0 and 255
- Example: **192.168.1.1** → that's Octet 1 . Octet 2 . Octet 3 . Octet 4
- IP addresses **can change** — they are assigned temporarily and can move to different devices
- But two devices on the same network can **never have the same IP at the same time**

Two types of IP address

Private IP

Used to identify a device inside your own network (home, office). Example: 192.168.1.77 — only meaningful inside your house.

Public IP

Used to identify your whole network on the Internet. Given by your ISP (Internet Service Provider — the company you pay for internet). Every device in your house shares the same public IP.

Real example: Two computers in a house. Computer A: private IP 192.168.1.77, public IP 86.157.52.21. Computer B: private IP 192.168.1.74, public IP 86.157.52.21. Same public IP — different private IPs. They talk to each other using private IPs. They talk to the Internet using the shared public IP.

IPv4 vs IPv6

→ **IPv4** = the original version. Supports ~4.29 billion addresses (2^{32}). We're running out.

→ **IPv6** = the new version. Supports 340 trillion+ addresses (2^{128}). Solves the shortage problem.

→ IPv4 looks like: 86.157.52.21 — IPv6 looks like: 2a00:22c4:a531:c500:425f:cce6:c36b:f64d

Why does this matter for cybersecurity? An IP address is how you trace where traffic is coming from or going to. Understanding private vs public IPs helps you determine if suspicious activity is internal (inside the network) or external (from the internet).

MAC addresses — the permanent identity

What is a MAC address?

→ MAC = **Media Access Control**. It's a permanent address burned into a device's network interface chip at the factory

→ Every device has one — it's like a fingerprint. Unlike an IP address, it doesn't change when you move networks

→ Format: 12 characters split in pairs by colons → example: a4:c3:f0:85:ac:2d

→ First 6 characters = **which company** made the network chip (vendor ID)

→ Last 6 characters = **unique number** for that specific device

Devices have two identifiers — remember this analogy

Like a human being

Your name = IP address (can change — you can change your name). Your fingerprints = MAC address (can never change — burned in at birth / at the factory). Both are used to identify you, but one is permanent and one isn't.

MAC spoofing — the security problem

→ MAC addresses **can be faked** — this is called **spoofing**

→ A device can pretend it has a different MAC address — tricking other devices into thinking it's someone else

Real-world attack example

A firewall is set to only allow traffic from the admin's MAC address. An attacker spoofs the admin's MAC address. The firewall now thinks it's talking to the admin — and lets the attacker through. This is why MAC-based security alone is not enough.

You see MAC address control in hotels and coffee shops — their guest Wi-Fi only works if your MAC is registered. Some charge per device using this system.

Ping — testing if a connection works

What is ping?

- Ping is a basic tool that tests if two devices can reach each other on a network
- Uses **ICMP packets** (Internet Control Message Protocol) — tiny messages designed just for testing connections
- Sends an **echo request** → waits for an **echo reply** → measures the time it took (in milliseconds)
- Syntax: **ping 192.168.1.254** or **ping google.com**

How to read ping output

Example: **ping 192.168.1.254**

6 packets transmitted, 6 received, 0% packet loss — average time = 4.16ms

"6 packets sent" = you pinged 6 times. "6 received" = all 6 came back. "0% packet loss" = perfect connection. "4.16ms" = average time for the round trip.

What ping tells you

- **Gets a reply** = the device exists on the network and is reachable
- **No reply** = device is down, unreachable, or blocking ICMP (some devices are configured to ignore pings)
- **High response time** (e.g. 500ms+) = connection is slow or unstable
- **Packet loss** = some of your pings didn't make it back — unreliable connection

Why it matters in cybersecurity: ping is the first thing you use to check if a device is alive on a network. Attackers use it during reconnaissance to map which IP addresses have active devices behind them.

LAN topologies — how networks are arranged

Topology = the shape or layout of a network. How devices are physically connected to each other.

Star topology

All devices connect to one central hub or switch. Like a star shape.

Advantages

Easy to add more devices (scalable). If one cable breaks, only that device goes down — not the whole network. Most common type today.

Disadvantages

More expensive — needs more cable and dedicated hardware. If the central hub/switch fails, every device loses connection.

Bus topology

All devices connect to a single cable (called the backbone). Like a tree branch with leaves.

Advantages

Cheap and easy to set up. Less cabling needed.

Disadvantages

Gets slow fast — all data shares one cable. Hard to troubleshoot. One cable break = entire network down. No backup.

Ring topology

Devices connect in a loop — data travels around the ring until it reaches the right device.

Advantages

Easy to troubleshoot — data only travels one direction. Less prone to bottlenecks. Less cabling than star.

Disadvantages

Slow — data may have to travel through many devices to reach its destination. One break in the loop = whole network fails.

Memory trick: Star = most reliable and common. Bus = cheapest but fragile (one cable = everything dies). Ring = middle ground, but one break still kills everything.

Switches and routers — the traffic directors

Switch

A switch connects multiple devices inside the same network

Devices plug into a switch's ports (4, 8, 16, 24, 32, or 64 ports). Common in businesses, schools, and offices.

→ A switch is **smart** — it knows which device is connected to which port. When it gets data, it sends it **only to the right device**, not to everyone.

→ A **hub** (the older, dumber version) sends data to every device — wasteful and slower.

Router

A router connects different networks together and passes data between them

The router in your house connects your private home network to the public Internet. The process of directing data between networks is called routing.

→ Routing = creating a path for data to travel from one network to another

→ Routers and switches can be connected to each other to create **redundancy** — multiple paths for data. If one path breaks, another is used. No downtime.

Switch vs Router — the one-line difference

Switch = directs traffic inside one network. Router = directs traffic between different networks. Your home router does both — it's actually a router + switch + Wi-Fi access point combined.

Subnetting — dividing a network into smaller pieces

What is subnetting?

→ Subnetting = splitting one big network into smaller mini-networks called **subnets**

→ Each subnet gets its own range of IP addresses and behaves somewhat independently

Easy way to remember it

Think of slicing a cake

There's only a certain amount of cake (IP addresses) to go around. Subnetting is deciding who gets which slice and how big each slice is. A business might give one slice to HR, one to Finance, one to Accounting — each department gets their own section of the network.

Three ways subnets use IP addresses

Network address

Identifies the network itself. Example: 192.168.1.0

Host address

Identifies each device on the network. Example: 192.168.1.100

Default gateway

The device that sends data out to other networks. Example: 192.168.1.254

Why subnetting matters in security

→ Subnetting lets you **separate** different parts of a network from each other

→ Example: A café has one subnet for staff and registers, another for the guest Wi-Fi. Customers can't see staff devices — even though they're in the same building on the same internet connection

Security benefit: if a hacker gets onto the guest Wi-Fi, they're isolated in that subnet. They can't reach the cash register or employee computers on the staff subnet.

ARP — how devices find each other on a network

What is ARP?

→ ARP = [Address Resolution Protocol](#)

→ Devices have two identifiers: an IP address and a MAC address. ARP is the technology that [links them together](#)

→ Every device keeps a small notebook called an [ARP cache](#) — a list of other devices' IP and MAC address pairings

How ARP works — step by step

1

[ARP Request](#) — device broadcasts to the whole network: "Hey everyone — who has IP address 192.168.1.10? What's your MAC address?"

2

[ARP Reply](#) — the device with that IP replies privately: "That's me — my MAC address is 18:AC:33:12:88:29"

3

The asking device saves that IP-to-MAC mapping in its ARP cache — so next time it doesn't have to ask again

Memory trick: ARP is like shouting "Who lives at number 10?" in a street. The person at number 10 shouts back "Me! I'm at number 10." Now you know which door to knock on directly.

Security note — ARP poisoning

Because ARP broadcasts to everyone on the network, attackers can send fake ARP replies to poison other devices' caches — making them think the attacker's MAC address belongs to the gateway. This lets them intercept all traffic. This is called an ARP poisoning or man-in-the-middle attack.

DHCP — how devices automatically get an IP address

What is DHCP?

→ DHCP = [Dynamic Host Configuration Protocol](#)

→ Instead of manually typing an IP address into every device, DHCP does it automatically

→ There's a [DHCP server](#) on the network (usually built into your router) that hands out IP addresses to devices that connect

The 4-step process — DORA

1

DHCP Discover

New device joins the network and broadcasts: "Is there a DHCP server here? I need an IP address!"

2

DHCP Offer

DHCP server replies: "Yes! You can have IP address 192.168.1.10 — want it?"

3

DHCP Request

Device replies: "Yes please! I'll take 192.168.1.10."

4

DHCP ACK (Acknowledgement)

Server confirms: "Done — that address is yours. You have it for the next 24 hours."

Memory trick: DORA — Discover, Offer, Request, Acknowledge. Like walking into a hotel: you arrive (discover) → desk offers you a room (offer) → you say yes (request) → they hand you the key (ACK).

Security note: Attackers can set up a fake DHCP server on a network — called a rogue DHCP server. If your device accidentally gets an IP address from the fake server instead of the real one, the attacker can redirect all your traffic through them.

Quick-reference summary

- ★ Network = devices connected. Private (home/business) or public (Internet)
- ★ Internet invented by Tim Berners-Lee in 1989 — built on ARPANET from the 1960s
- ★ IP address = device's address. 4 octets (0–255). Can change. Private vs Public
- ★ IPv4 = ~4.3 billion addresses. IPv6 = 340 trillion+ addresses (solves the shortage)
- ★ MAC address = permanent fingerprint. 12 hex characters. First 6 = manufacturer, last 6 = unique ID
- ★ MAC can be spoofed — don't rely on MAC alone for security
- ★ Star topology = most common, most reliable. Bus = cheap but fragile. Ring = one direction, one break kills it
- ★ Switch = directs traffic inside one network (smart — sends to the right port only)
- ★ Router = connects different networks together and routes data between them
- ★ Subnetting = slicing a network into smaller parts. Security benefit: isolates departments from each other
- ★ ARP = links IP addresses to MAC addresses. ARP cache = the notebook of saved pairings
- ★ DHCP = automatically assigns IP addresses. Process: Discover → Offer → Request → ACK (DORA)
- ★ Ping = tests if a device is reachable. Uses ICMP echo request + reply. Measures time in milliseconds